

Laboratory Session

Acoustic Full Waveform Inversion (FWI)

Master 2 – Acoustic Imaging

1 Introduction: Nonlinear Inverse Problems

In seismic imaging, we aim to reconstruct the physical properties of the subsurface (wave velocity, density, etc.) from measured wavefields at the surface. This is a fundamentally **nonlinear inverse problem**, since the relation between the model parameters $\mathbf{m}$ (e.g., "velocity") and the observed data $\mathbf{d}$ is governed by the wave equation.

1.1 Formulation of the Inverse Problem

The objective is to minimize a **cost (misfit) function**:

$$J(\mathbf{m}) = \frac{1}{2} \|\mathbf{d}_{obs} - \mathbf{d}_{calc}(\mathbf{m})\|^2 \quad (1)$$

where $\mathbf{d}_{obs}$ are the observed data and $\mathbf{d}_{calc}(\mathbf{m})$ are the modeled data obtained by solving the forward problem for a given model $\mathbf{m}$.

The nonlinearity of the problem arises because $\mathbf{d}_{calc}$ depends on $\mathbf{m}$ through the wave equation. This leads to several challenges:

- Existence of multiple local minima,
- Strong sensitivity to the initial model,

The method known as **Full Waveform Inversion (FWI)** exploits the complete wavefield information (both phase and amplitude) to iteratively recover the parameters of the medium using local optimization techniques.

2 Governing Equations

2.1 Forward Problem: Helmholtz Equation

In the frequency domain, acoustic wave propagation is governed by the **Helmholtz equation**:

$$\nabla^2 p(\mathbf{x}, \omega) + \frac{\omega^2}{c^2(\mathbf{x})} p(\mathbf{x}, \omega) = -s(\mathbf{x}, \omega) \quad (2)$$

where $p(\mathbf{x}, \omega)$ is the acoustic pressure at position $\mathbf{x}$ and frequency ω , $c(\mathbf{x})$ is the wave velocity, and $s(\mathbf{x}, \omega)$ is the source term.

Introducing the squared slowness $m(\mathbf{x}) = 1/c^2(\mathbf{x})$, the equation becomes:

$$\nabla^2 p(\mathbf{x}, \omega) + \omega^2 m(\mathbf{x}) p(\mathbf{x}, \omega) = -s(\mathbf{x}, \omega) \quad (3)$$

The forward problem consists of computing the pressure field p for a known source s and model $m(\mathbf{x})$.

2.2 Inverse Problem: Full Waveform Inversion

For each source s_i and frequency ω the Eq. (1) can be rewrite as:

$$J(m) = \sum_i \sum_{\omega} \sum_r \frac{1}{2} |p_{obs}(\mathbf{x}_r, \omega, s_i) - p_{calc}(\mathbf{x}_r, \omega, s_i; m)|^2 \quad (4)$$

where $\mathbf{x}_r$ are receiver positions.

The minimization FWI proceeds iteratively according to:

$$m^{(k+1)} = m^{(k)} - \alpha^{(k)} \nabla_m J$$

where $\alpha^{(k)}$ is the step length, determined by line search or empirical scaling.

2.3 Gradient Computation: The Adjoint-State Method

Computing $\nabla_m J$ directly would require differentiating the wavefield with respect to the model parameters, which is computationally expensive. The **adjoint-state method** provides an elegant and efficient alternative.

Let p be the forward wavefield satisfying the Helmholtz equation, and p^* the adjoint wavefield satisfying:

$$\nabla^2 p^*(\mathbf{x}, \omega) + \omega^2 m(\mathbf{x}) p^*(\mathbf{x}, \omega) = \Delta p(\mathbf{x}_r, \omega) \quad (5)$$

where $\Delta p = p_{calc} - p_{obs}$ is the data residual injected at the receiver locations.

Then, the gradient of the cost function with respect to the model is:

$$\nabla_m J = \sum_i \sum_{\omega} \text{Re} [\omega^2 p(\mathbf{x}, \omega, s_i) \cdot p^*(\mathbf{x}, \omega, s_i)] \quad (6)$$

Physical interpretation: The gradient corresponds to the zero-lag correlation between the forward-propagated wavefield (from the sources) and the back-propagated residual field (from the receivers). It highlights the regions where changes in the model would reduce the data misfit.

2.4 Sensitivity Kernels and Resolution

The quantity

$$K(\mathbf{x}) = \omega^2 p(\mathbf{x}, \omega) \cdot p^*(\mathbf{x}, \omega) \quad (7)$$

is called the **sensitivity kernel**. It describes how a perturbation in the model at point $\mathbf{x}$ affects the misfit.

Typical kernels have a “banana–doughnut” shape:

- Maximum sensitivity lies along the paths connecting sources and receivers.
- The resolution is proportional to the local wavelength $\lambda = c/f$.
- Low frequencies probe broad structures, while high frequencies enhance fine details but are more sensitive to noise and modeling errors.

2.5 Optimization and Regularization

FWI is a local optimization problem and often uses gradient-based methods (steepest descent, conjugate gradient, L-BFGS). Regularization techniques are essential to stabilize the inversion:

- **Tikhonov regularization:** penalizes large parameter variations,
- **Multi-scale approach:** start with low frequencies to recover the background structure, then progressively include higher frequencies.

3 Exercises

3.1 Part 1: Understanding the Forward Problem

Exercise 1.1 – Forward Modeling

Modify the provided velocity model by $\Delta c = \pm 10\%$ (for example). Generate the corresponding synthetic data in frequency and in time.

Questions:

- a) Do you observe differences in arrival times? Why?
- b) How are the amplitudes and phase affected?
- c) Hint: $f(t) = A_n \cos(\omega_n t + \phi_n)$

Exercise 1.2 – Frequency Sensitivity Analysis

Compute synthetic data for three frequencies: $f_1 > f_2 > f_3$ Hz.

Questions:

- a) What relationship do you observe between frequency and wavelength?
- b) How does this influence the spatial resolution of anomalies?

3.2 Part 2: Gradients and Sensitivity Kernels

Exercise 2.1 – Sensitivity Kernels

Visualize the sensitivity kernels $K(\mathbf{x})$ for different source–receiver pairs (at least two).

Questions:

- a) What is the typical shape of a kernel (e.g., “banana” pattern)?
- b) How does the acquisition geometry affect sensitivity coverage?

Exercise 2.2 – Gradient Computation

Using the provided code, compute and visualize the gradient of the cost function for a given source and frequency.

Questions:

- a) Where is the gradient strongest? Interpret physically.
- b) Compare the gradient with the true location of the anomaly.

3.3 Part 3: Practical Inversion

Exercise 3.1 – Inversion with Perfect Synthetic Data

Create a simple “true” model. Generate corresponding synthetic data, then start from a smooth initial model (constant or with a vertical gradient). Run the FWI and analyze the convergence.

Questions:

- a) Does the inverted model correctly recover the anomalies?
- b) Plot the evolution of the cost function with iterations.
- c) Are there any artifacts? Where and why?

Exercise 3.2 – Multi-Scale Strategy

This exercise illustrates the importance of multi-scale inversion in avoiding local minima. Perform three inversions:

1. **Low frequencies only:** $f \in [f_1, f_2]$ Hz
2. **High frequencies only:** $f \in [f_3, f_4]$ Hz
3. **Progressive strategy:** start with low frequencies and gradually add higher ones

Questions:

- a) Compare the three results. Which one yields the best reconstructed model?
- b) Why does the high-frequency-only inversion fail or diverge?
- c) What is the dominant wavelength in each case, and how does it relate to the achieved resolution?

Exercise 3.3 – Noise Robustness

Add Gaussian white noise to the synthetic data (SNR = 20 dB, then 10 dB).

Questions:

- a) How does the inversion quality degrade with increasing noise?
- b) Suggest regularization strategies to improve robustness.

3.4 Part 4: Bonus Questions

Exercise 4.1 – Acquisition Geometry

Compare inversion results for:

- Dense configuration: X_1 sources, Y_1 receivers
- Sparse configuration: X_2 sources, Y_2 receivers

Question: How does angular coverage affect inversion resolution?

Exercise 4.2 – Influence of the Initial Model

Test several initial models (homogeneous, linear gradient, random perturbations).

Question: In which cases does the inversion converge to the true model?